

HEART DISEASE ANALYSIS

Project Report

1. INTRODUCTION

1.1 Project Overview

Project Overview – Heart Disease Analysis

Heart Disease Analysis is a data visualization and analytics project designed to analyze patient health records and identify the key factors associated with heart disease. The project transforms raw healthcare data into meaningful insights using Tableau dashboards, enabling users to explore trends, compare patient characteristics, and support data-driven healthcare decisions.

Interactive dashboards and stories provide clear visualizations, including KPIs, trend charts, comparison graphs, and distribution analyses. These visual insights help healthcare professionals, researchers, policymakers, and students better understand disease patterns, identify high-risk patient groups, and make informed decisions based on data.

Objectives:

- Analyze patient health data to identify major risk factors for heart disease.
- Build interactive dashboards for easy exploration of healthcare data.
- Present meaningful insights through KPIs and visual analytics.
- Support data-driven decision-making for healthcare professionals and researchers.
- Demonstrate practical skills in data preprocessing, visualization, and business intelligence using Tableau.

This project showcases the application of business intelligence and data visualization techniques in the healthcare domain, making complex medical data easier to interpret and enabling more effective analysis of heart disease trends and risk factors.

1.2 Purpose

The purpose of the Heart Disease Analysis project is to analyze patient health data and identify the major factors that contribute to heart disease using data visualization and business intelligence techniques. By transforming raw healthcare data into interactive dashboards, the project helps users understand patterns, trends, and relationships among various medical attributes.

This project aims to provide clear, data-driven insights that support informed decision-making for healthcare professionals, researchers, and policymakers. It enables users to identify high-risk patient groups, compare health indicators, and monitor key performance metrics through intuitive visualizations.

Additionally, the project demonstrates the practical application of Tableau in data preprocessing, dashboard development, and analytical reporting. It serves as a learning tool for students and data analysts while showcasing how business intelligence can improve healthcare analysis and decision-making.

2. IDEATION PHASE

2.1 Problem Statement

Heart disease is one of the leading causes of death worldwide. Although healthcare organizations collect large amounts of patient data, it is often difficult to analyze and interpret using traditional methods. This makes it challenging to identify risk factors, recognize disease patterns, and support timely decision-making.

The Heart Disease Analysis Dashboard addresses this problem by transforming raw patient data into interactive visualizations using Tableau. The dashboard enables users to identify key risk factors, analyze patient trends, and gain data-driven insights that support better healthcare decisions.

<https://docs.google.com/document/d/1OmGJ2L2LfYqRZAwb3rkd31-k8yfjPfUGEYqDqREsC58/edit?usp=sharing>

2.2 Empathy Map Canvas

The **Empathy Map Canvas** is a user-centered design tool used to understand the needs, challenges, behaviors, and expectations of the target users. It helps identify what users **see, hear, think, feel, say, and do**, along with their **pains** and **gains**, enabling the development of solutions that effectively address real-world problems.

For the **Heart Disease Analysis Dashboard**, the empathy canvas focuses on healthcare professionals, hospital administrators, researchers, and public health officials. It highlights their need for accurate, interactive, and easy-to-understand data visualizations to support faster decision-making. By understanding users' challenges, such as analyzing large healthcare datasets and identifying disease trends, the project delivers an intuitive Tableau dashboard that provides meaningful insights, improves data accessibility, and supports informed healthcare decisions.

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 Empathy Map Canvas.docx - Google Docs

2.3 Brainstorming

For the Heart Disease Analysis Dashboard, brainstorming sessions were conducted to identify the major challenges in analyzing healthcare data and to determine the most effective way to present meaningful insights. Various visualization techniques, KPIs, dashboard layouts, filters, and storytelling approaches were discussed. Based on these discussions, an interactive Tableau dashboard was designed to analyze heart disease trends, identify key risk factors such as age, gender, smoking, diabetes, and BMI, and support data-driven decision-making for healthcare professionals and researchers. The brainstorming process ensured

that the final solution was user-friendly, informative, and aligned with the project's objectives.

 BrainstormingTemplate - Google Docs

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

For the **Heart Disease Analysis Dashboard**, the customer journey begins with healthcare professionals or researchers recognizing the need for better analysis of heart disease data. They access the Tableau dashboard, explore interactive visualizations, apply filters to analyze specific patient groups, and interpret insights related to risk factors such as age, gender, smoking, diabetes, BMI, and stroke. These insights support informed clinical decisions, healthcare planning, and preventive measures. By simplifying complex healthcare data into intuitive visualizations, the dashboard enhances user experience, reduces analysis time, and enables faster, data-driven decision-making.

 Customer Journey Map - Example - Google Docs

<https://drive.google.com/file/d/1r9nHA3jKMeIill6Gq4wISrOeIuPo2sT0/view?usp=sharing>

3.2 Solution Requirement

The proposed solution requires a reliable and well-structured heart disease dataset that can be cleaned, preprocessed, and analyzed using Tableau. The system should support interactive dashboards with KPI cards, filters, and visualizations to help users explore heart disease trends and risk factors such as age, gender, smoking, diabetes, BMI, race, and general health. It should include calculated fields for meaningful analysis, enable easy navigation through story points, and present insights in a clear and user-friendly format. The dashboard must provide accurate, interactive, and real-time visual analysis to assist healthcare professionals, researchers, and decision-makers in identifying high-risk groups, understanding disease patterns, and making informed healthcare decisions.

 Solution Requirements.docx - Google Docs

https://drive.google.com/file/d/1hwOtCFzELbIGPUnpigPf7q-FLdM_qZ5d/view?usp=sharing

3.3 Data Flow Diagram

1. Heart disease data is collected from a CSV/Excel dataset.
2. The dataset is imported into Tableau.
3. Data preprocessing is performed by cleaning data, removing duplicates, handling missing values, and validating records.
4. Tableau creates calculated fields and establishes the data model.
5. Interactive dashboards, KPIs, filters, and story points are developed.
6. Healthcare professionals, researchers, and hospital administrators interact with the dashboard to analyze trends and make informed decisions.

 Data Flow Diagrams and User Stories.docx - Google Docs

https://drive.google.com/file/d/1Z4yppqbkoLXUsatGIVL2_qnOFg78qMuC/view?usp=sharing

3.4 Technology Stack

The project uses a Heart Disease dataset in CSV/Excel format as the primary data source. The data is cleaned and preprocessed using Microsoft Excel before being imported into Tableau Desktop, where interactive dashboards, KPI cards, filters, charts, and story points are created. The completed project is published on Tableau Public for online access and maintained in a GitHub repository for version control and collaboration. Project documentation is prepared using Microsoft Word, while Microsoft PowerPoint is used for presenting the project's objectives, methodology, and findings.

 [Technology Stack - Template.docx - Google Docs](#)

https://drive.google.com/file/d/1JKUqNeXzcqMiA2cdL4_72uqlLg9sRaHO/view?usp=sharing

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Problem–Solution Fit ensures that the proposed solution effectively addresses the real problems faced by its target users. In this project, healthcare professionals and researchers often face challenges in analyzing large volumes of heart disease data, identifying key risk factors, and extracting meaningful insights from complex datasets. Traditional methods of data analysis are time-consuming and make it difficult to recognize patterns that support timely decision-making.

To address these challenges, the Heart Disease Analysis Dashboard was developed using Tableau. The dashboard transforms raw healthcare data into interactive visualizations, KPI cards, and story-based analyses that help users explore trends, compare patient groups, and identify major risk factors such as age, gender, smoking, diabetes, BMI, and stroke. By providing an intuitive and interactive platform, the solution enables faster analysis, supports evidence-based healthcare decisions, and improves the overall understanding of heart disease patterns.

 [Proposed Solution Template - Google Docs](#)

<https://drive.google.com/file/d/1HFjMDObB8W3GcOuHyXb69VYhRFa3mL4d/view?usp=sharing>

4.2 Proposed Solution

The proposed solution is an interactive Heart Disease Analysis Dashboard developed using Tableau to transform raw healthcare data into meaningful visual insights. The system collects heart disease data, performs preprocessing to remove inconsistencies, and presents the information through interactive dashboards, KPI cards, charts, and story points. Users can apply filters based on factors such as age, gender, race, smoking status, diabetes, BMI, general health, and stroke history to analyze patient trends and identify high-risk groups.

https://drive.google.com/file/d/1I8ITLiXarVb1gdv8bmZkT_Lqn7IE6uCh/view?usp=sharing

4.3 Solution Architecture

The solution architecture begins with a **Heart Disease dataset** stored in CSV or Excel format. The dataset undergoes **data preprocessing**, including removal of duplicates, handling of missing values, data cleaning, and validation to ensure quality and consistency. The cleaned data is then imported into **Tableau Desktop**, where calculated fields and data models are created for analysis.

 Solution Architecture - Google Docs

<https://docs.google.com/document/d/1cTokpnf8nXtXuYqB7M89Qx0IJ-pm-Blj-KUYMdzJzus/edit?usp=sharing>

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Project planning is the process of defining the project's objectives, scope, timeline, resources, and deliverables to ensure its successful completion. For the Heart Disease Analysis Dashboard, project planning was carried out using an Agile approach by dividing the work into multiple sprints. Each sprint focused on specific tasks such as requirement analysis, data collection, preprocessing, dashboard development, testing, deployment, and documentation. This structured approach helped monitor progress, manage resources efficiently, and ensure the project was completed within the planned schedule.

 Project Planning Template - Google Docs

<https://drive.google.com/file/d/1UMog0sR2CoPJXi-cgsslnd5pYz2uHzt0/view?usp=sharing>

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

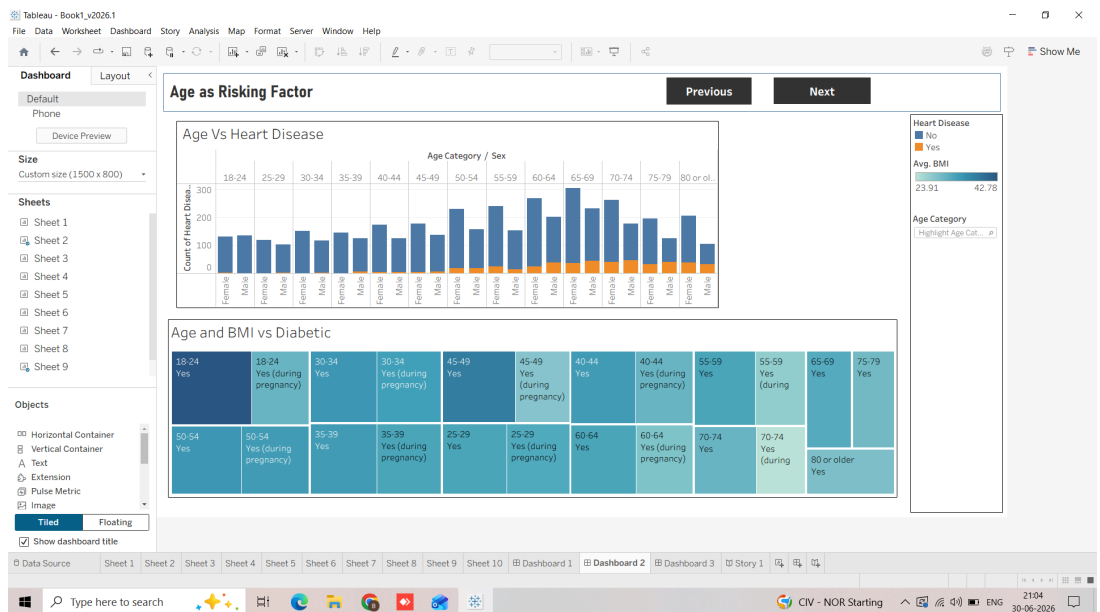
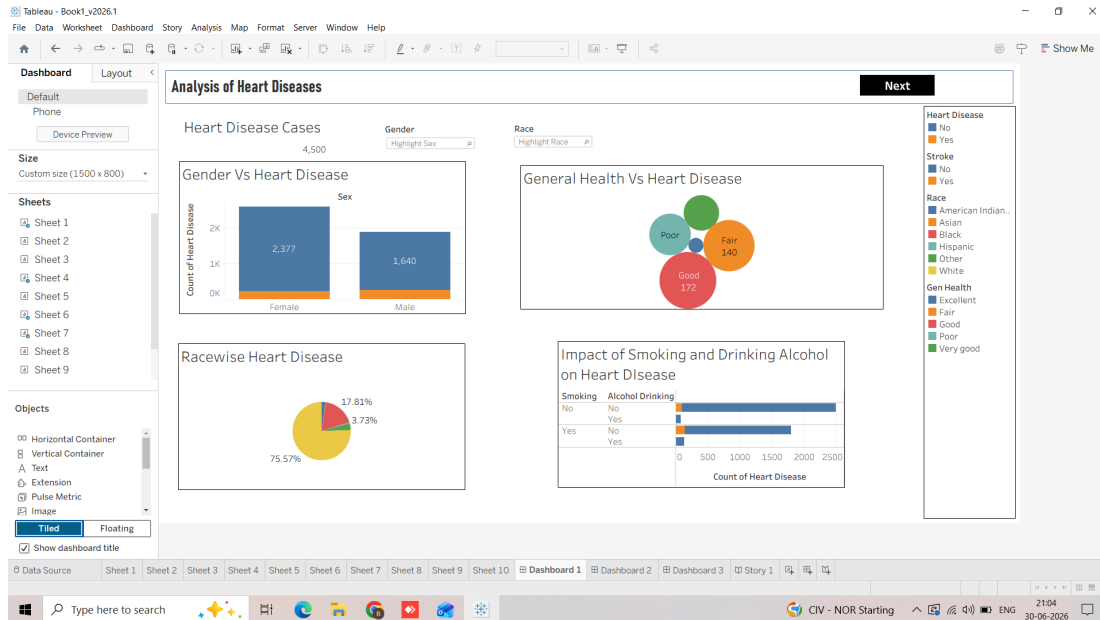
Performance testing was conducted to evaluate the functionality, responsiveness, and accuracy of the Heart Disease Analysis Dashboard developed using Tableau. The testing process ensured that the dashboard correctly processed the dataset, displayed accurate visualizations, and responded efficiently to user interactions such as filtering and drill-down analysis. Data preprocessing steps, including handling missing values, removing duplicates, and validating data types, ensured the reliability of the analysis. The dashboard was tested for KPI accuracy, calculated fields, chart rendering, filter performance, and story navigation. Results confirmed that the dashboard loads successfully, provides accurate insights, and enables smooth interaction without noticeable delays. The successful completion of performance testing demonstrates that the system is reliable, user-friendly, and capable of supporting healthcare professionals and researchers in analyzing heart disease data effectively.

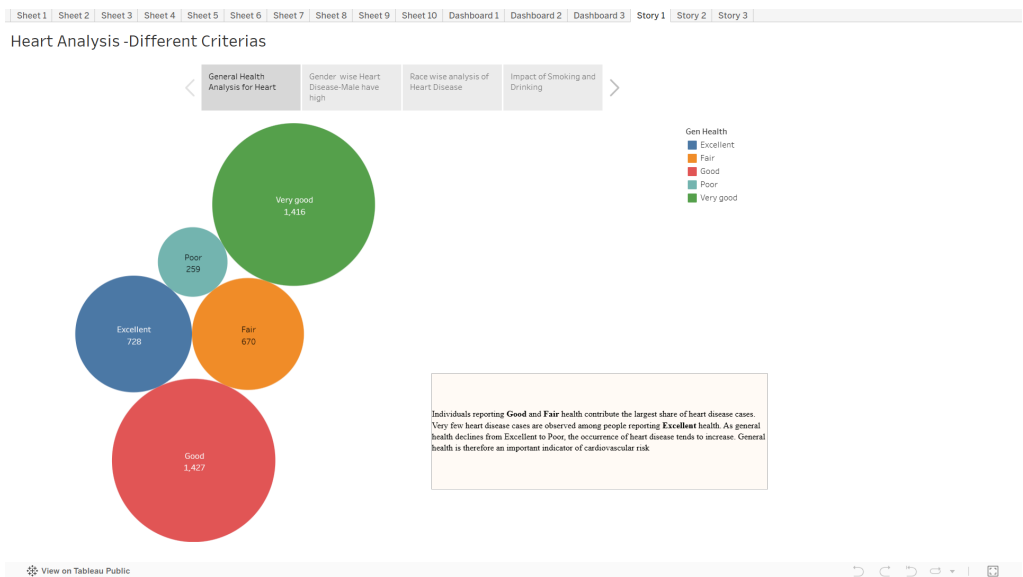
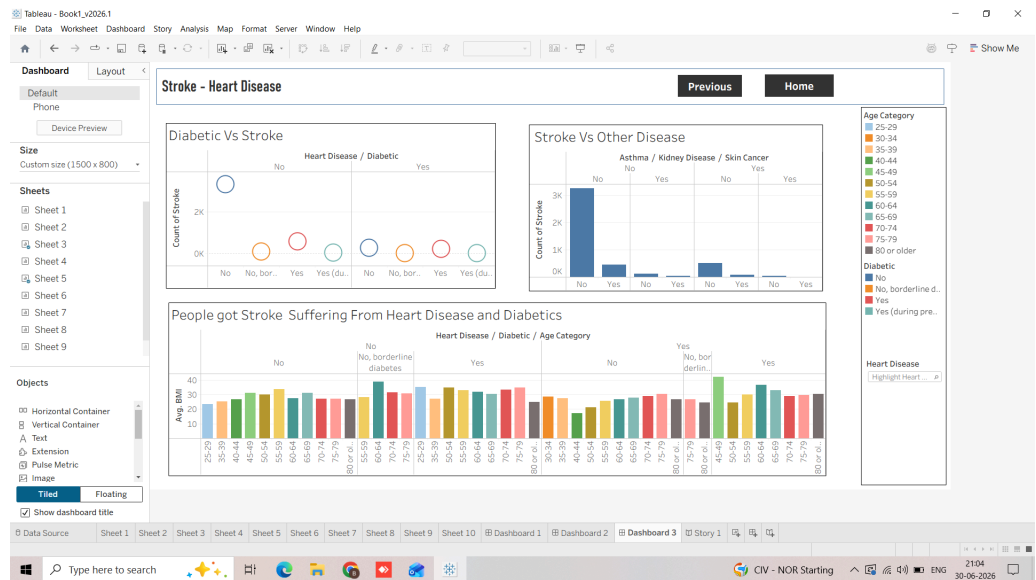
https://drive.google.com/file/d/1g1Cq0V02v6Er3N9ZWRKdoO_o9DAwSwON/view?usp=sharing

 Project Performance Template - Google Docs

7. RESULTS

7.1 Output Screenshots





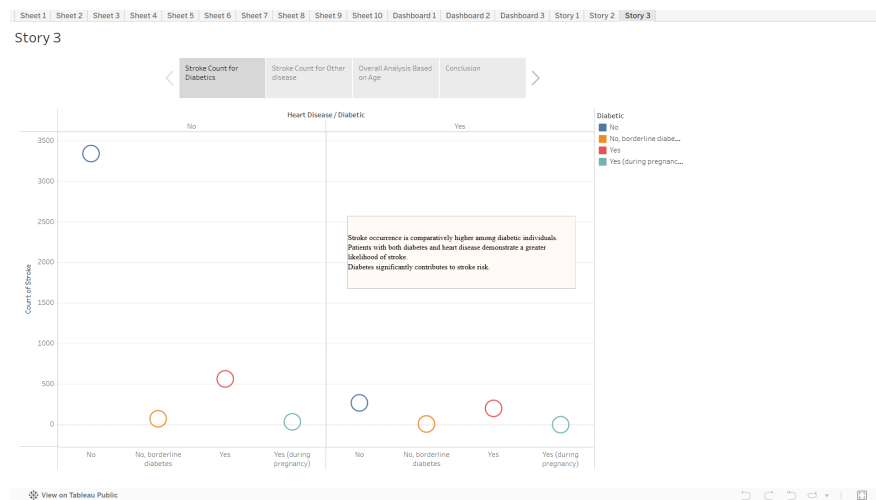
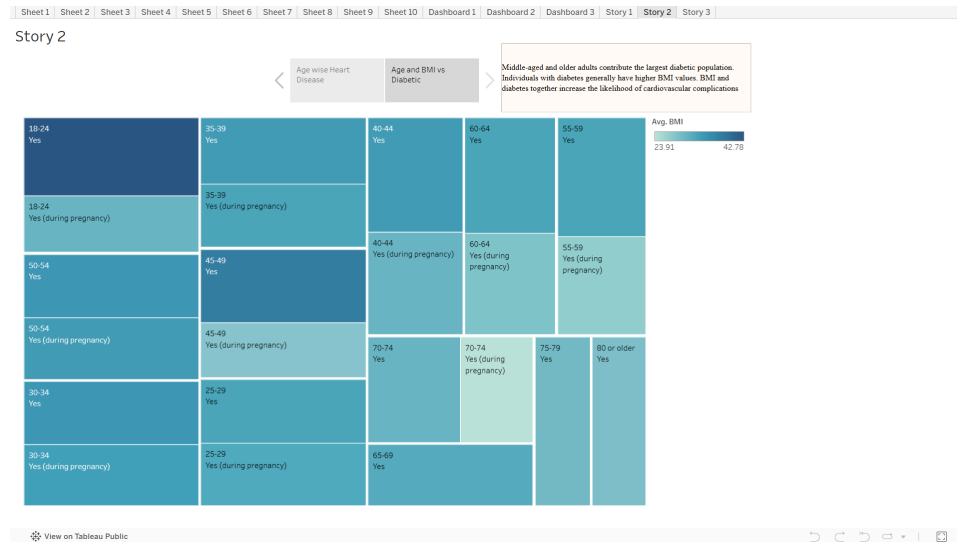


TABLEAU PUBLIC LINK:

https://public.tableau.com/views/HeartDiseaseAnalysis_17829802569990/Story3?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

8. ADVANTAGES & DISADVANTAGES

ADVANTAGES:

1. **Interactive Visualization:** Presents complex heart disease data through easy-to-understand charts, graphs, and dashboards.
2. **Data-Driven Decision Making:** Helps healthcare professionals identify trends and risk factors for informed decisions.
3. **User-Friendly Interface:** Interactive filters and dashboards allow users to explore data with ease.
4. **Comprehensive Analysis:** Examines multiple risk factors such as age, gender, BMI, smoking,

diabetes, stroke, and general health.

5. **Supports Preventive Healthcare:** Helps identify high-risk patient groups for early intervention.
6. **Improved Reporting:** Generates clear visual reports that are easy to interpret and present.

DISADVANTAGES:

1. **Dependent on Data Quality:** Incorrect or incomplete data can lead to inaccurate insights.
2. **Requires Tableau Knowledge:** Users need basic familiarity with Tableau to modify or extend the dashboard.
3. **Static Dataset:** Results depend on the available dataset and do not update automatically unless new data is imported.
4. **Internet Requirement for Tableau Public:** Accessing the published dashboard online requires an internet connection.

9. CONCLUSION

The Heart Disease Analysis Dashboard developed using Tableau successfully transforms raw healthcare data into meaningful and interactive visual insights. By analyzing key risk factors such as age, gender, smoking, diabetes, BMI, general health, stroke history, and race, the dashboard enables users to identify patterns and understand the factors associated with heart disease. Interactive dashboards, KPI cards, filters, and story points simplify complex medical data, making it easier for healthcare professionals, researchers, and hospital administrators to perform efficient analysis and support data-driven decision-making.

The project demonstrates the effectiveness of data visualization in healthcare by reducing the time required for manual analysis and improving the accessibility of critical health information. Overall, the Heart Disease Analysis Dashboard provides a user-friendly and scalable solution for monitoring heart disease trends, identifying high-risk groups, and supporting preventive healthcare initiatives. Future enhancements, such as real-time data integration and predictive analytics using machine learning, can further improve the system's capabilities and contribute to better healthcare outcomes.

10. FUTURE SCOPE

The Heart Disease Analysis Dashboard can be further enhanced by incorporating advanced technologies and additional healthcare data sources. Future improvements can make the system more intelligent, scalable, and beneficial for healthcare professionals and patients.

- Integrate real-time patient data from hospital databases and Electronic Health Records (EHR) for continuous monitoring.
- Implement Machine Learning models to predict heart disease risk based on patient health parameters.
- Incorporate Artificial Intelligence to provide personalized risk assessments and healthcare recommendations.
- Develop a web-based or mobile application for easy access to the dashboard anytime and anywhere.

11. APPENDIX

DATASET LINK: [Heart_new2 \(1\).csv.xlsx](#)

GitHub : <https://github.com/HimaVani123/Heart-Disease-Analysis--Tableau>